

LIÈGE université
Sciences Appliquées

Frequency control

ELEC0447 - Analysis of electric power and energy systems

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Lecture Roadmap

1. Physics of Power Balance & Frequency Dynamics
2. Primary Frequency Control
3. Secondary Frequency Control (AGC / aFRR)
4. Tertiary Control, Economic Dispatch & OPF

References and Source Material

- Lecture notes of **Prof. Thierry Van Cutsem** (ULiège)
- **Mohan, Ned.** *Electric Power Systems: A First Course*. John Wiley & Sons, 2012 (Chapter 12, summarized by Prof. Louis Wehenkel).
- **Kundur, Prabhashankar.** *Power System Stability and Control*. McGraw-Hill, 1994.
- ENTSO-E Operational Handbook and Balancing Guidelines.

Pedagogical Design

This deck is organized into **4 recording modules** ($\approx$ 20 min each) spanning physics, local droop, centralized AGC, and economic dispatch / OPF.

1. Physics of Power Balance & Frequency Dynamics

Why Control Power Systems?

- **Strict technical tolerance regions:**
  - Voltage magnitude V_n: 1 ± 0.1 p.u. across European grids.
  - **Grid frequency f_n: 50 ± 0.2 Hz in steady state.**
  - Current limits: $I \leq I_{\max}$ on lines and transformers.
- **Consequences of severe frequency deviations:**
  - Turbine blade resonance fatigue and mechanical failure.
  - Under-/over-frequency relay trips (< 47.5 Hz or > 51.5 Hz).
  - Cascading collapse of interconnected synchronous areas.

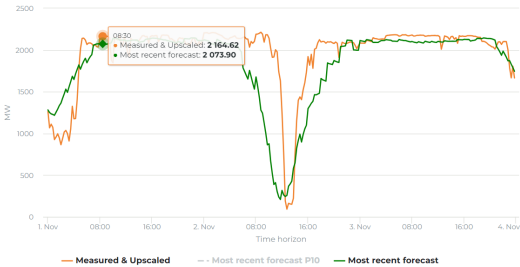
Golden Principle

Electrical energy cannot be stored directly in large quantities on the AC grid; **generation must match demand at every single microsecond**.

Exogenous Threats: Wind Storms & Fast Gradients

Case Study: Storm Ciarán (2 Nov 2023, Belgium)

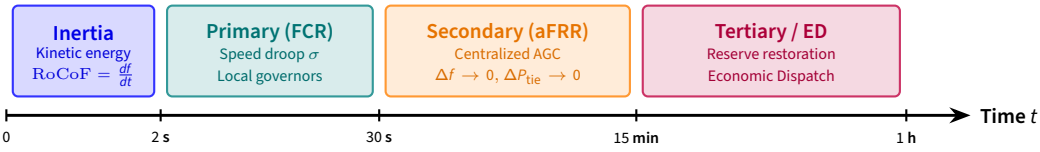
- High-speed gale winds reached offshore cut-out thresholds ($>$ 25 m/s).
- Belgian offshore wind plunged from **4.5 GW to 0.5 GW** in under 30 minutes!
- Such massive active power deficits threaten frequency stability unless fast reserves respond immediately.



Source: Elia Transmission Belgium (`https://www.elia.be`)

The Multi-Scale Frequency Control Hierarchy

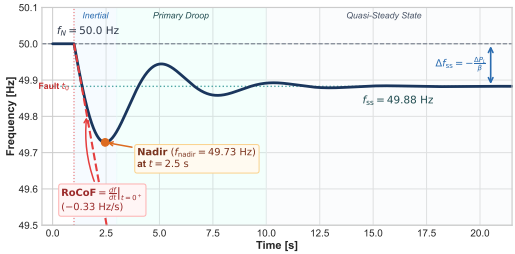
Frequency restoration is coordinated across distinct physical and market layers:



Frequency Dynamic Signature: The Frequency Nadir

Dynamic Stages Following a Generation Loss (ΔP_L):

1. **Inertial Response ($0 - 2$ s):** Stored kinetic energy buffers the shortfall; initial deceleration is governed by the **Rate of Change of Frequency (RoCoF)**.
2. **Frequency Nadir (f_{nadir} at $2 - 6$ s):** Point where governor mechanical ramp equals the power deficit ($df/dt = 0$).
3. **Primary Settling ($10 - 30$ s):** Quasi-steady state stabilized at $f_{ss} < f_N$ by droop.
4. **Secondary Recovery ($1 - 15$ min):** AGC integral action drives frequency back to 50.0 Hz (not shown).



Control Resources: Levers for Frequency Stability

Which resources directly act on frequency?

- **Synchronous generator mechanical power:** Main lever (valves/gates in thermal, gas, hydro units).
- **Battery Energy Storage Systems (BESS):** Ultra-fast synthetic inertia and FCR (response within < 200 ms).
- **Fast start-up peakers (OCGT, hydro pumps):** Secondary / tertiary replenishment.
- **Inverter curtailment / over-frequency control:** Rapid downward regulation for solar PV and wind.
- **Demand-side response & UFLS:** Controlled load disconnection in emergency states.

Non-Frequency Levers

Field excitation (E_f), transformer taps (OLTC), and shunt capacitors control **voltage** ($Q - V$), not frequency!

System Frequency Evolution: Theory, Example, and Intuition

Before exploring multi-level frequency control, let us build intuition from a simplified physical power system model.

Consider an interconnected grid with loads and synchronous generators:

- $p_e(t)$ is the total electrical power absorbed by loads (including network transmission losses and generator conversion losses).
  - By Kirchhoff's laws, this is the exact total active power generators must inject at every instant!
- $p_m(t)$ is the total mechanical power delivered to generator turbine shafts.

Fundamental Question: **How does system frequency $f(t)$ (and rotor mechanical speed) evolve over time?**

Physical Intuition

Electrical demand $p_e(t)$ is dictated by consumers. Mechanical power $p_m(t)$ is controlled by turbine valves. Any instantaneous imbalance accelerates or decelerates generator rotors!

The Swing Equation

Consider a synchronous area operating near nominal frequency f_N. Group all synchronous machines into a single equivalent rotor with moment of inertia J.

Newton's second rotational law relates rotor acceleration α to torque:

$$J\alpha = T_m - T_e$$

Since $p = T\omega$ and $\omega = 2\pi f$, we obtain:

$$J\frac{df}{dt} = \frac{p_m - p_e}{4\pi^2 f}$$

Adding generator mechanical damping D_g (friction, windage, damper windings):

$$J\frac{df}{dt} = \frac{p_m - p_e}{4\pi^2 f} - D_g(f - f_N)$$

Kinetic Buffer

When $p_e > p_m$, missing power is drawn directly from rotating kinetic energy $E_k = \frac{1}{2}J\omega^2$, causing $f(t)$ to decline.

Inertia Constant of a Generator

The physical moment of inertia J_i of generator i is linked to its rated power $P_{\text{max},i}$ and dimensionless inertia constant H_i:

$$J_i = \frac{2P_{\text{max},i}H_i}{4\pi^2 f_N^2} \quad [\text{kg} \cdot \text{m}^2]$$

where f_N is nominal grid frequency (50 or 60 Hz).

- H_i (in seconds) represents the duration the machine can supply its rated power $P_{\text{max},i}$ solely from stored kinetic energy.
- Typical values: $H \approx 2 - 4$ s for hydro, $H \approx 4 - 7$ s for thermal units.
- In the swing equation, the total system inertia is $J = \sum_i J_i$.

System Inertia

In a coherent synchronous grid, all rotors swing in synchronism, acting as one giant equivalent flywheel of total inertia $J = \sum_i J_i$.

What Happens If... (Interactive Questions)

1. Power Imbalance (Sudden Load Increase ΔP_L)?

Rotors immediately decelerate ($df/dt < 0$). Without governor intervention, frequency drops continuously until damping bounds it or the system collapses!

2. Effect of Lower Inertia ($J \downarrow$)?

Steeper initial slope (higher $\mathrm{RoCoF} = |df/dt|$), leading to a faster and deeper frequency nadir.

3. Effect of Damping (D_g)?

Provides natural resistive braking proportional to frequency deviation, moderating the decline.

Simulation Verification

Verify these 3 physical effects in the interactive notebook:

▸ Notebook: Cases 1–3

- **Case 1:** No governor ($\sigma \to \infty$)
- **Case 2:** Low inertia ($J/10$)
- **Case 3:** Zero damping ($D_g = 0$)

Let's Add Generator Reaction to Frequency Change

Why Inertia and Damping Are Not Enough:

- Kinetic energy buffering is purely passive and temporary; it cannot restore power balance.
- Damping alone leaves an unacceptably large frequency deviation far below statutory limits (± 200 mHz).
- Therefore, generators must actively adjust their mechanical power output $p_m(t)$ in response to frequency changes!

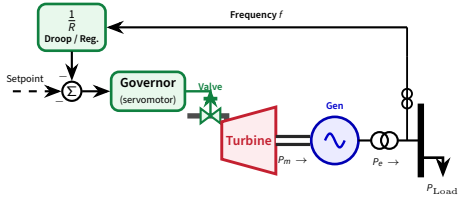
The Primary Control Loop:

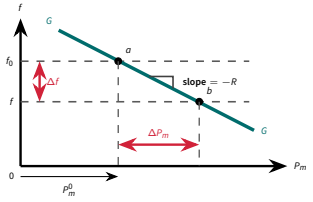
- This fast automatic adjustment is **primary frequency control**.
- Let us analyze how a power plant's mechanical valve and speed governor are modeled.

Control Necessity

To prevent continuous frequency decay, every major generator is equipped with a local speed governor.

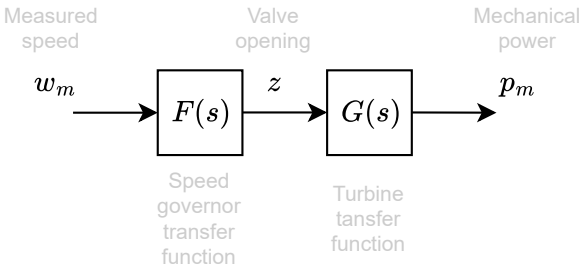
Speed Governor of a Synchronous Generator





- Rotor speed is continuously measured (or electrical frequency as a proxy).
- When frequency drops below f_N, the governor opens the valve to increase steam/gas/water mass flow.
- In steady state, frequency and mechanical power follow a **linear droop** characteristic.
- This relationship determines the generator's active power response during an imbalance.
- **Proportional control principle:** to trigger a power adjustment, frequency must deviate!

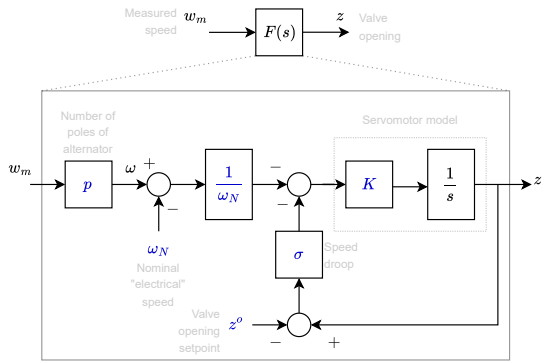
Model of a Power Plant (Mechanical Part)



Mechanical Drive Components:

- ω_m is the mechanical rotor speed (linked to electrical frequency ω by number of pole pairs).
- $z \in [0, 1]$ is the fractional opening of turbine control valves.
- $G(s)$ models the turbine response (converting valve opening z into shaft torque/power p_m).
- We focus primarily on the **speed governor** block that commands z.

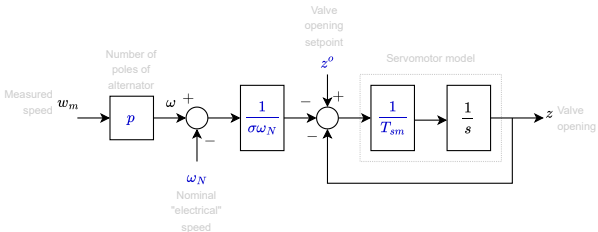
Speed Governor Transfer Function



Governor Signal Flow:

- Rotor speed is scaled and compared to reference ω_N to generate a speed error signal.
- z^o sets the desired power production setpoint at nominal frequency.
- In the feedback branch, valve position z is compared to z^o and scaled by droop σ.
- Net error drives the hydraulic servomotor (integrator with gain K) to adjust valve position z.

Equivalent Block Diagram & Governor Transfer Function



Servomotor Transfer Function:

- Defining servomotor time constant $T_{sm} = \frac{1}{K\sigma}$ (typically $0.2 - 1.5$ s).

- The closed-loop transfer function is:

$$z(s) = \frac{1}{1 + sT_{sm}} \left(z^o - \frac{\omega - \omega_N}{\sigma\omega_N} \right)$$

- In steady state ($s \to 0$):

$$z = z^o - \frac{\omega - \omega_N}{\sigma\omega_N}$$

Steady-State Characteristic of a Turbine-Governor Set

In steady state ($s \to 0$):

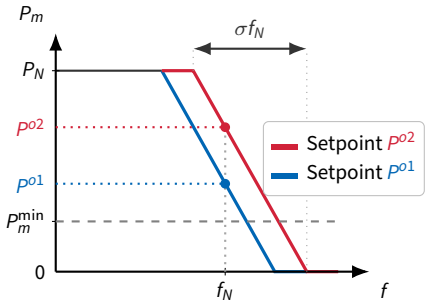
- $p_m = G(0)z$ where $G(0)$ is the static turbine gain.
- At full valve opening ($z = 1$), turbine reaches nominal power $P_N = G(0) \cdot 1 \implies p_m = P_N z$.
- Substituting $z = z^o - \frac{\omega - \omega_N}{\sigma \omega_N}$:

$$p_m = P_N z^o - \frac{P_N}{\sigma} \frac{f - f_N}{f_N}$$

- Defining scheduled power setpoint $P^o = P_N z^o$:

$$\boxed{p_m = P^o - \frac{P_N}{\sigma} \frac{f - f_N}{f_N}}$$

subject to physical limits $p_m \in [P_{\min}, P_{\max}]$.



Time-Domain Turbine-Governor Model & Limits

Transforming back to the time domain yields the first-order dynamic ODE for mechanical power:

$$T_{sm}\frac{dp_m}{dt} = P^o - p_m - \frac{P_N}{\sigma}\frac{f - f_N}{f_N}$$

with typical parameters $T_{sm} \approx 0.5 - 2$ s and droop $\sigma \approx 4\% - 6\%$.

Physical Generator Boundaries:

$$p_m(t) \in [P_{\min}, P_{\max}]$$

- If p_m hits $P_{\max}$, upward reserve is exhausted (saturation).
- If p_m hits $P_{\min}$, downward curtailment capability is lost.

Simulation Verification

Explore droop variations and saturation limits in the notebook:

▸ Notebook: Cases 4–7

- **Case 4:** Standard 4% droop
- **Case 5:** Lighter 8% droop
- **Case 6:** Saturation at $P_{\max}$
- **Case 7:** Downward limit $P_{\min}$

About the Speed Droop σ (Le Statisme)

Speed Droop σ Definition

The speed droop is the ratio between relative frequency deviation and relative power deviation:

$$\sigma = \left| \frac{\Delta f / f_N}{\Delta P_m / P_N} \right| = \left| \frac{\Delta \omega / \omega_N}{\Delta P_m / P_N} \right|$$

- A frequency drop $\Delta f = \sigma f_N = 0.04 \times 50 = 2$ Hz mobilizes 100% nominal capacity ($\Delta P_m = P_N$).
- For normal primary activation band $\Delta f = -200$ mHz (-0.4%), a 4% droop mobilizes $\frac{0.004}{0.04} P_N = 10\% P_N$.
- If $\sigma \to \infty$, the generator operates at constant power and does not participate in frequency control.

Proportional Action

Because speed droop is a proportional controller:

- It inherently leaves a **steady-state error** ($\Delta f \neq 0$).
- But this error is the **common signal** that allows multiple plants to share the effort without communicating!

The Power Consumed by the Load Is Frequency Dependent

In reality, load power demand varies with grid frequency:

$$p_e(f) = P_e^o \left(1 + D_l(f - f_N)\right)$$

where P_e^o is nominal consumption, and $D_l \approx 1\% - 2\%$ per Hz ($0.01 - 0.02$ Hz^{-1}) is the load frequency sensitivity.

Physical Origin:

- Induction motors driving fans, pumps, and compressors slow down as frequency falls, drawing less mechanical power.
- Resistive loads (heaters, lighting) remain frequency-independent.
- Natural self-regulation provides an inherent stabilizing damping effect!

Stabilizing Feedback

When frequency drops, total load automatically decreases, reducing the active power deficit!

The Impact of Renewable Generation

Assume non-synchronous renewable generation $p_{PV}(t)$ is added to the system.

Which equation does it impact?

$$J\frac{df}{dt} = \frac{p_m - p_e + p_{PV}}{4\pi^2 f} - D_g(f - f_N)$$

Coupled Physical Model Summary:

$$T_{sm}\frac{dp_m}{dt} = P^o - p_m - \frac{P_N}{\sigma}\frac{f - f_N}{f_N}, \quad p_m \in [P_{\min}, P_{\max}]$$

$$p_e = P_e^o\,(1 + D_l(f - f_N))$$

Consequence of RES Growth:

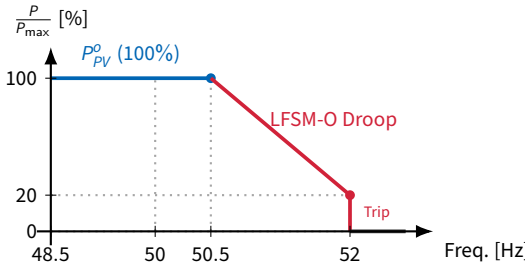
- When PV displaces synchronous generators, total inertia J drops.
- An unexpected solar surge without control causes overfrequency.

Simulation Verification

Simulate a sudden $+30$ MW solar surge without curtailment in the notebook:

▸ Notebook: Case 8

PV Inverter Modeling & Overfrequency Curtailment (LFSM-O)



Piecewise Mathematical Model:

$$p_{PV}(f) = \begin{cases} P^o_{PV} & \text{if } f < 50.5 \text{ Hz} \\ P^o_{PV}\left[1 - \frac{0.8}{1.5}(f - 50.5)\right] & \text{if } 50.5 \leq f < 52.0 \text{ Hz} \\ 0 & \text{if } f \geq 52.0 \text{ Hz} \end{cases}$$

Grid Code Mandates (ENTSO-E RfG):

- Inverters cannot provide upward reserve unless pre-curtailed below MPPT.
- **LFSM-O (Overfrequency):** For $f \geq 50.2$ or 50.5 Hz, active power must curtail automatically with a fast droop ($2\% - 12\%$).
- **LFSM-U (Underfrequency):** Pre-curtailed units ramp up power when $f < 49.8$ Hz.

▸ Notebook: Cases 9 & 10

2. Primary Frequency Control

Objective and Assumptions

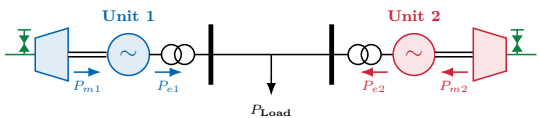
Analyze the steady-state characteristics of primary control across an interconnected system under key assumptions:

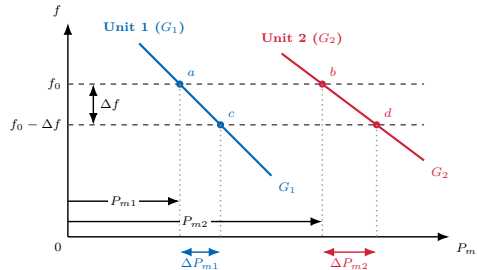
- The post-disturbance system has settled to quasi-steady state ($10 - 30$ s).
- All synchronous machines rotate at the same electrical frequency ($f_1 = f_2 = \cdots = f$).
- All participating generators are active within their operational bounds.
- Lossless transmission network ($\sum p_{mi} = p_e$).
- Mechanical turbine power is fully converted into electrical power.
- Load depends on frequency via sensitivity D_l: $p_e(f) = P_e^o(1 + D_l \Delta f)$.
- Initial operation is perfectly balanced at nominal frequency ($f = f_N$).

Core Question

How is a sudden load disturbance autonomously distributed among multiple participating plants without telecommunication?

Parallel Operation: Two Generators Sharing a Load Increase





Autonomous Load Sharing:

- Load increase $\Delta P_L \implies$ rotors decelerate, frequency drops ($\Delta f < 0$).
- Both governors open valves autonomously along droop curves (no communication).
- At steady state, power balance is restored:

$$\Delta P_{m1} + \Delta P_{m2} = \Delta P_L$$

Share of Power Variation Among n Generators

Aggregating droop curves of n machines ($\Delta f = f - f_N$):

$$p_m = \sum_{i=1}^{n} p_{mi} = \sum_{i=1}^{n} \left(P_i^o - \frac{\Delta f}{f_N} \frac{P_{Ni}}{\sigma_i} \right)$$

Power balance with demand disturbance ΔP_e:

$$\sum_{i=1}^{n} \left(P_i^o - \frac{\Delta f}{f_N} \frac{P_{Ni}}{\sigma_i} \right) = P_e^o (1 + D_l \Delta f) + \Delta P_e$$

Using initial balance ($\sum P_i^o = P_e^o$) and rearranging:

$$-\frac{\Delta f}{f_N} \sum_{i=1}^{n} \frac{P_{Ni}}{\sigma_i} = P_e^o D_l \Delta f + \Delta P_e$$

$$\Longrightarrow \quad \boxed{\Delta f = -\frac{\Delta P_e}{\beta}}$$

Key Result

The steady-state frequency drop is directly proportional to the disturbance ΔP_e and inversely proportional to system stiffness β.

System Stiffness (β) and Generator Participation

Composite Frequency Response Characteristic (β)

$$\beta = \frac{1}{f_N} \sum_{i=1}^{n} \frac{P_{Ni}}{\sigma_i} + D_l P_e^o \quad \text{[MW/Hz]}$$

Also called *network power frequency characteristic* or *stiffness of system* (en français: *énergie régulante*).

- Continental Europe: $\beta \approx 15\,000 - 25\,000$ MW/Hz ($\approx 1500 - 2500$ MW per 0.1 Hz).

Variation of Power of j-th Generator

$$\Delta P_{mj} = -\frac{\Delta f}{f_N} \frac{P_{Nj}}{\sigma_j} = \boxed{\frac{\Delta P_e}{\beta} \frac{P_{Nj}}{\sigma_j f_N}}$$

Load Sharing Properties:

- **Size rule:** Larger rated power $P_{Nj} \implies$ larger contribution.
- **Droop rule:** Smaller droop σ_j (steeper curve) $\implies$ larger contribution.
- **System scale:** More participating generators $\implies$ larger β, smaller Δf.

Primary Reserve (FCR) Market Structure

Operational Headroom & Eligibility:

- Only certified units participate in Frequency Containment Reserve (FCR).
- Generators must maintain headroom: $p_m \in [P_{\min} + R, P_{\max} - R]$.
- Base-load nuclear units rarely ramp; renewables offer mostly downward reserve.

ENTSO-E Regulatory Framework:

- Reference incident sizing: $N - 2 = 3000$ MW in Continental Europe.
- Symmetrical full activation band: ± 200 mHz (49.8 $-$ 50.2 Hz).
- Deployment within **30 seconds**; endurance $\geq$ 15 minutes.
- Capacity procured through competitive regional market auctions.

End of Part 2

Shortcomings of Primary Control:

1. Residual error: $\Delta f \neq 0$.
2. Unscheduled cross-border tie-line flows!

$\Longrightarrow$ Motivates Secondary Control (AGC)!

3. Secondary Frequency Control (AGC / aFRR)

Why Do We Need Secondary Frequency Control?

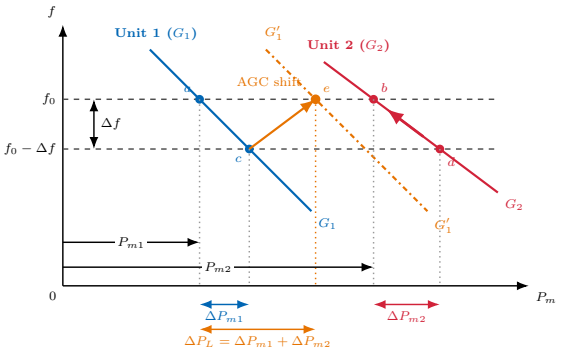
Primary control achieves power balance, but leaves three critical shortcomings:

1. **Persistent Frequency Offset:** $\Delta f = -\Delta P_e/\beta \neq 0$. Clocks and synchronized equipment drift over time.
2. **Unscheduled Cross-Border Flows:** Neighboring countries help absorb the imbalance, violating commercial tie-line contracts ($\Delta P_{\text{tie}} \neq 0$).
3. **Depleted Primary Reserves:** Synchronous generators are left running away from nominal setpoints, leaving zero headroom for a second fault!

Mission of AGC

Automatic Generation Control (AGC / aFRR) deploys within $0.5 - 15$ min to drive both $\Delta f \to 0$ and $\Delta P_{\text{tie}} \to 0$, restoring primary reserves.

Secondary Control: Governor Setpoint Adjustment



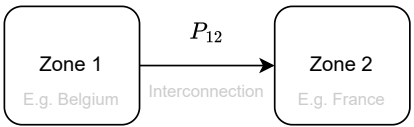
Secondary Action (AGC):

- **Setpoint Shift:** AGC adjusts Unit 1 speed changer ($G_1 \rightarrow G_1'$).
- **Restoration:** Frequency returns to nominal ($f \rightarrow f_0$).
- **Reserve Recovery:** Unit 2 rides G_2 back to point b, fully recovering its primary reserve.
- **Result:** Unit 1 covers the entire load increase ΔP_L (point e).

Two Interconnected Areas: Physical Model

Consider two synchronous areas interconnected by AC tie-lines:

- Area 1: $P_{m1} = P_{e1} + P_{12}$
- Area 2: $P_{m2} = P_{e2} - P_{12}$



where P_{12} is the net export flow from Area 1 to Area 2.
Disturbance Scenario: Sudden load increase ΔP_{e1} in Area 1.
Primary droop responds across **both** areas:

$$-\beta_1 \Delta f = \Delta P_{e1} + \Delta P_{12}$$
$$-\beta_2 \Delta f = -\Delta P_{12}$$

Analysis of the Two-Area Interconnection

Summing the two equations yields total frequency deviation:

$$\Delta f = -\frac{\Delta P_{e1}}{\beta_1 + \beta_2}$$

Substituting Δf back into Area 2 equation gives tie-line flow deviation:

$$\Delta P_{12} = -\frac{\beta_2}{\beta_1 + \beta_2}\Delta P_{e1} < 0$$

- Power flows from Area 2 into Area 1 to support the deficit.
- Area 2 generators are supporting Area 1 at their own expense!
- Secondary control in Area 1 must increase internal production so Area 2 can return to its scheduled exchange P_{12}^0.

Responsibility Principle

The area where the imbalance originated must activate secondary reserve to bring its net interchange back to schedule!

Exercise: Two-Area Primary Response

System Data: Two interconnected synchronous areas:

- Area 1: $\beta_1 = 10\,000$ MW/Hz, Area 2: $\beta_2 = 15\,000$ MW/Hz
- Initial state: $f_0 = 50.000$ Hz, scheduled tie-line exchange $P_{12}^0 = 0$ MW
- Disturbance: sudden load step in Area 1 of $\Delta P_{e1} = +1\,000$ MW ($\Delta P_{e2} = 0$).

Questions:

1. Compute the new quasi-steady-state frequency deviation Δf.
2. Compute the primary power response ($\Delta P_{m1}, \Delta P_{m2}$) in each area.
3. Determine the unscheduled tie-line power flow deviation ΔP_{12} ($1 \to 2$). Which area supports which?

Solution: Two-Area Primary Response

System Data:

- Area 1: $\beta_1 = 10\,000$ MW/Hz, Area 2: $\beta_2 = 15\,000$ MW/Hz
- Disturbance: sudden load step in Area 1 of $\Delta P_{e1} = +1\,000$ MW.

1. Frequency Deviation (Δf):

$$\Delta f = -\frac{\Delta P_{e1}}{\beta_1 + \beta_2} = -\frac{1\,000}{25\,000} = -\mathbf{0.040}\ \text{Hz} \quad (-40\ \text{mHz})$$

2. Primary Response per Area ($\Delta P_{mi} = -\beta_i \Delta f$):

- Area 1: $\Delta P_{m1} = -10\,000 \times (-0.04) = +\mathbf{400}$ MW (40%)
- Area 2: $\Delta P_{m2} = -15\,000 \times (-0.04) = +\mathbf{600}$ MW (60%)

3. Unscheduled Tie-Line Flow Deviation ($1 \to 2$):

$$\Delta P_{12} = \beta_2 \Delta f = 15\,000 \times (-0.04) = -\mathbf{600}\ \text{MW}$$

$\Longrightarrow$ **600 MW flows from Area 2 into Area 1** to help cover the deficit!

Physical Interpretation

- Area 2 automatically covers **60%** of Area 1's deficit through local droop!
- Tie-line flow drops by 600 MW, violating commercial exchange schedules.
- Area 2's primary reserves are depleted for Area 1's event.

The Area Control Error (ACE)

To localize the imbalance without inter-TSO real-time communication, each area i computes its **Area Control Error (ACE)**:

$$\mathrm{ACE}_i = \Delta P_{\mathrm{tie},i} + \lambda_i \Delta f$$

where:

- $\Delta P_{\mathrm{tie},i} = P_{\mathrm{tie},i} - P^0_{\mathrm{tie},i}$ is net tie-line power flow deviation (export positive).
- $\Delta f = f - f_N$ is global frequency deviation.
- $\lambda_i > 0$ is the **frequency bias factor** [MW/Hz].

Sign Convention

If $\mathrm{ACE}_i < 0$, Area i has an internal active power deficit and must increase generation.

Exercise: Area Control Error (ACE)

Context from previous primary response state:

- $\Delta f = -0.040$ Hz, $\Delta P_{12} = -600$ MW (net export $\Delta P_{\text{tie},1} = -600$ MW, $\Delta P_{\text{tie},2} = +600$ MW)
- Recall definition: $\text{ACE}_i = \Delta P_{\text{tie},i} + \lambda_i \Delta f$

Questions:

1. Express $\text{ACE}_1(\lambda_1)$ and $\text{ACE}_2(\lambda_2)$ with frequency bias factors λ_1, λ_2 as parameters.
2. What value must λ_2 take so that Area 2 (unaffected area) does not deploy secondary control ($\text{ACE}_2 = 0$)?
3. If $\lambda_1 = \beta_1 = 10\,000$ MW/Hz, evaluate ACE_1. How does it relate to the initial disturbance ΔP_{e1}?

Solution: Area Control Error (ACE)

From the previous primary response state:

- $\Delta f = -0.040$ Hz, $\Delta P_{12} = -600$ MW (Area 2 exports to Area 1)
- $\Delta P_{\text{tie},1} = \Delta P_{12} = -600$ MW, $\Delta P_{\text{tie},2} = -\Delta P_{12} = +600$ MW

1. Expressing ACE with λ as a parameter:

$$\text{ACE}_1(\lambda_1) = \Delta P_{\text{tie},1} + \lambda_1 \Delta f = -\mathbf{600} - \mathbf{0.040}\,\lambda_\mathbf{1}$$
$$\text{ACE}_2(\lambda_2) = \Delta P_{\text{tie},2} + \lambda_2 \Delta f = +\mathbf{600} - \mathbf{0.040}\,\lambda_\mathbf{2}$$

2. Condition for Area 2 (healthy) not to intervene ($\text{ACE}_2 = 0$):

$$600 - 0.040\,\lambda_2 = 0 \iff \lambda_2 = \frac{600}{0.040} = \mathbf{15\,000}\ \text{MW/Hz} = \beta_2!$$

3. Resulting ACE in Area 1 (with $\lambda_1 = \beta_1 = 10\,000$ MW/Hz):

$$\text{ACE}_1 = -600 - 0.040 \times 10\,000 = -\mathbf{1\,000}\ \text{MW} = -\Delta P_{e1}$$

$\Longrightarrow$ Area 1 AGC ramps up exactly 1 000 MW of secondary reserves!

Impact of Tuning λ_2

- **Matched ($\lambda_2 = 15\,000$):** $\text{ACE}_2 = 0$ MW $\Longrightarrow$ zero secondary intervention.
- **Under-biased ($\lambda_2 = 10\,000$):** $\text{ACE}_2 = +200$ MW > 0 $\Longrightarrow$ Area 2 backs off generation!
- **Over-biased ($\lambda_2 = 20\,000$):** $\text{ACE}_2 = -200$ MW < 0 $\Longrightarrow$ Area 2 takes on Area 1's deficit!

Optimal Tuning of the Bias Factor λ_i

What happens if we choose $\lambda_2 = \beta_2$ for Area 2 (where no fault occurred)?

$$\begin{aligned} \text{ACE}_2 &= \Delta P_{\text{tie},2} + \lambda_2 \Delta f \\ &= -\Delta P_{12} + \beta_2 \left(-\frac{\Delta P_{e1}}{\beta_1 + \beta_2} \right) \\ &= -\left(-\frac{\beta_2}{\beta_1 + \beta_2} \Delta P_{e1} \right) - \frac{\beta_2}{\beta_1 + \beta_2} \Delta P_{e1} = 0! \end{aligned}$$

Fundamental Property of Tie-Line Bias Control

If each TSO tunes $\lambda_i = \beta_i$, **only the faulted area measures a non-zero ACE**! Healthy areas do not incorrectly react.

Decentralized Intelligence

By matching λ_i to stiffness β_i, multinational grids self-discriminate local vs external imbalances purely from local boundary flows!

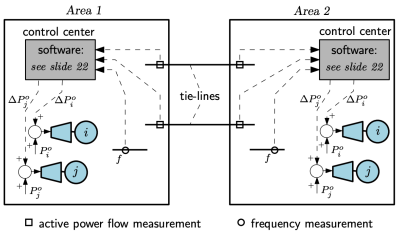
Centralized AGC Architecture

Centralized Dispatch Loop at the TSO:

- SCADA samples tie-line flows and grid frequency every $1 - 4$ s.
- ACE is fed into a Proportional-Integral (PI) controller:

$$\Delta P^{o}_{\text{area}} = -K_p \text{ACE} - K_i \int \text{ACE}\, dt$$

- In practice, $K_p \approx 0$ (pure integral control) to prevent fighting primary droop!
- When $\text{ACE} = 0 \implies \Delta f = 0$ and $\Delta P_{\text{tie}} = 0$.



Participation Factors & Dynamic Ramp Constraints

The total area correction ΔP^o_{area} is distributed among contracted aFRR units:

$$P^o_k \leftarrow P^o_k + \rho_k \Delta P^o_{\text{area}} \quad \text{with} \quad \sum_k \rho_k = 1$$

- ρ_k: participation factor of generator k.
- **Unit ramp rates limits must be respected:**
  - Hydro units: up to $100\%\, P_N/\text{min}$ (extremely agile).
  - Thermal / CCGT units: $2\% - 6\%\, P_N/\text{min}$.
  - Batteries: instantaneous response (< 1 s).
- Anti-windup algorithms prevent integral saturation when units hit $P_{\max}$.

Setpoints vs Output

AGC adjusts governor **power setpoints** P^o_k, leaving the primary droop feedback loop fully intact to handle fast fluctuations!

European Balancing Landscape: PICASSO

The PICASSO Platform (ENTSO-E):

- *Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation*.
- Optimizes the activation of aFRR bids across all participating European TSOs.
- Computes common merit-order activation every **4 seconds**.
- **Imbalance Netting:** If Belgium is short while Germany is long, their imbalances cancel out through cross-border capacity without activating expensive reserves!

End of Part 3

Summary: AGC restores frequency to 50 Hz and tie-lines to schedule. But who restores secondary reserve? **Tertiary control & Economic Dispatch**!

4. Tertiary Control, Economic Dispatch & OPF

Tertiary Control: Manual Restoration (mFRR & RR)

Objectives of Tertiary Control (15 min – 1 h):

- When a major outage persists, secondary reserves (aFRR) remain depleted.
- TSOs activate **Manual Frequency Restoration Reserve (mFRR)** via manual dispatch instructions.
- **Replacement Reserves (RR):** Start-up of slower thermal units or hydro pumped storage (e.g. Coo in Belgium) to relieve mFRR.
- **MARI platform:** Pan-European market for mFRR bid exchange.
- Once reserves are restored, unit operating setpoints are re-optimized via **Economic Dispatch**.

Time Scale

Tertiary control bridges real-time physical system balancing with intraday and day-ahead wholesale electricity markets!

The Classical Economic Dispatch Problem

Optimization Problem Statement: Find active generation setpoints $\{P_i\}_{i=1}^{n}$ that minimize total cost while meeting demand P_L:

$$\begin{aligned} \min_{\{P_i\}} \quad & \sum_{i=1}^{n} C_i(P_i) \\ \text{s.t.} \quad & \sum_{i=1}^{n} P_i = P_L + P_{\text{loss}} \quad \perp \quad \lambda \\ & \underline{P}_i \le P_i \le \overline{P}_i \quad \perp \quad \underline{\mu}_i, \overline{\mu}_i, \quad \forall i = 1, \ldots, n \end{aligned}$$

where $C_i(P_i) = a_i P_i^2 + b_i P_i + c_i$ is typically quadratic.

Notation Convention

- **Blue:** Primal decision variables (P_i).
- **Red:** Dual variables / multipliers ($\lambda, \underline{\mu}_i, \overline{\mu}_i$).
- **Black:** Fixed grid parameters ($P_L, \underline{P}_i, \overline{P}_i$).
- $\perp$: Complementarity / dual association with constraint.

Analytical Solution: Equal Incremental Cost Rule

Formulating the Lagrangian (neglecting losses for clarity):

$$\mathcal{L}(P_i, \lambda) = \sum_{i=1}^{n} C_i(P_i) - \lambda \left(\sum_{i=1}^{n} P_i - P_L \right)$$

First-order optimality condition ($\partial \mathcal{L} / \partial P_i = 0$ for unconstrained units):

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \cdots = \frac{dC_n}{dP_n} = \lambda$$

Merit Order Principle

All unconstrained generators must operate at the **exact same incremental marginal cost** λ [EUR/MWh], known as the **System Marginal Price (SMP)**!

KKT Boundary Cases

- If $dC_i/dP_i > \lambda \implies P_i = \underline{P}_i$ ($\underline{\mu}_i > 0$)
- If $dC_i/dP_i < \lambda \implies P_i = \overline{P}_i$ ($\overline{\mu}_i > 0$)

Units are stacked in ascending order of marginal cost!

Case Study: Belgian Power System Generation Mix

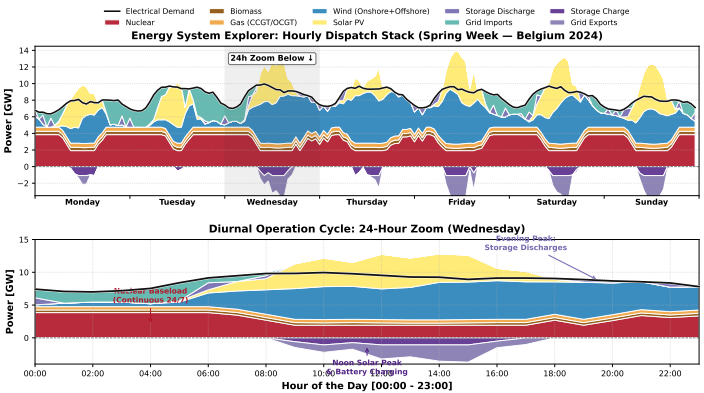
Structure of the Belgian Grid (ENVT3065):

- Annual demand: $\approx$ 85 TWh/year; peak load: $7 - 13$ GW.
- **Zero marginal cost:** Solar PV ($\sim$ 8 GW peak), Offshore wind (2.3 GW), Onshore wind ($\sim$ 3 GW).
- **Baseload nuclear:** Doel 4 and Tihange 3 ($\sim$ 2 GW remaining post-extension, marginal cost $\approx 10 - 15$ EUR/MWh).
- **Mid-merit CCGT gas:** Combined Cycle Gas Turbines ($\sim$ 6 GW, marginal cost $60 - 90$ EUR/MWh).
- **Peaking OCGT / Diesel:** Open cycle turbines ($> 150 - 250$ EUR/MWh).
- **Hydro Storage:** Coo-Trois-Ponts pumped hydro (1.08 GW, 5.8 GWh).

Belgian Merit Order

Renewables dispatch first $\rightarrow$ Nuclear baseload $\rightarrow$ CCGT gas $\rightarrow$ Peakers / imports set the marginal price λ in peak hours.

Belgian Generation Dispatch in Action



Operational Dynamics

- **Top pane:** Generation stack (nuclear, CCGT, wind, solar).
- **Middle pane:** Coo pumped-storage cycling (pumping off-peak, generating on-peak).
- **Bottom pane:** Hourly System Marginal Price $\lambda(t)$ [EUR/MWh].
- High RES output depresses λ towards zero (merit-order effect).

Limitations of "Copper Plate" Economic Dispatch

Why does Economic Dispatch fail in transmission grids?

- Economic dispatch assumes power flows instantaneously and without impedance from any generator to any load.
- In reality, electrical power routes strictly according to **Kirchhoff's Laws** and branch impedances $\mathbf{Y}_{\text{bus}}$!
- Dispatching cheap northern offshore wind to supply southern industry can easily overload intermediate transmission corridors ($I_{ij} > I_{ij}^{\max}$).
- Voltage bounds ($0.95 \leq V_k \leq 1.05$ p.u.) and reactive power capabilities ($Q_{\min} \leq Q_i \leq Q_{\max}$) are completely ignored!

Grid Congestion

When transmission lines saturate, TSOs must deviate from economic merit order: **preventive redispatch** or **Optimal Power Flow**!

Optimal Power Flow (OPF): The Full Formulation I

The Full AC Optimal Power Flow (AC-OPF):

$$
\begin{aligned}
\min_{\mathbf{P_G},\mathbf{Q_G},\mathbf{V},\boldsymbol{\theta}} \quad & \sum_{i\in\mathcal{G}} C_i(P_{Gi}) \\
\text{s.t.} \quad & P_{Gi} - P_{Li} = V_i \sum_{j\in\mathcal{N}} V_j(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \quad \perp \quad \lambda_{P,i}, \quad \forall i \in \mathcal{N} \\
& Q_{Gi} - Q_{Li} = V_i \sum_{j\in\mathcal{N}} V_j(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) \quad \perp \quad \lambda_{Q,i}, \quad \forall i \in \mathcal{N} \\
& \underline{P}_{Gi} \le P_{Gi} \le \overline{P}_{Gi} \quad \perp \quad \underline{\mu}_{P,i}, \overline{\mu}_{P,i}, \quad \forall i \in \mathcal{G} \\
& \underline{Q}_{Gi} \le Q_{Gi} \le \overline{Q}_{Gi} \quad \perp \quad \underline{\mu}_{Q,i}, \overline{\mu}_{Q,i}, \quad \forall i \in \mathcal{G} \\
& V_i^{\min} \le V_i \le V_i^{\max} \quad \perp \quad \underline{\mu}_{V,i}, \overline{\mu}_{V,i}, \quad \forall i \in \mathcal{N} \\
& |S_{ij}(\mathbf{V},\boldsymbol{\theta})| \le S_{ij}^{\max} \quad \perp \quad \mu_{S,ij}, \quad \forall (i,j) \in \mathcal{L}
\end{aligned}
$$

Optimal Power Flow (OPF): The Full Formulation II

Key Insights into OPF:

- **Mathematical Nature:** Large-scale, non-convex, non-linear optimization program.
- **DC-OPF Approximation:** For fast clearing, assume $V_i \approx 1$ p.u., line resistances $R \ll X$, and small angle differences ($\sin \theta_{ij} \approx \theta_i - \theta_j$):

$$\min_{\mathbf{P}_G, \boldsymbol{\theta}} \sum_{i \in \mathcal{G}} C_i(P_{Gi}) \quad \text{s.t.} \quad \mathbf{P}_G - \mathbf{P}_L = \mathbf{B}\boldsymbol{\theta} \perp \boldsymbol{\lambda}, \quad |B_{ij}(\theta_i - \theta_j)| \leq P_{ij}^{\max} \perp \mu_{ij}$$

  Solvable as a convex Quadratic or Linear Program (QP / LP)!

- **Locational Marginal Prices (LMP):** The dual variables λ_i at each bus reveal local nodal prices reflecting energy cost + transmission congestion losses.

Synthesis: The Full Frequency Hierarchy

Layer	Time Scale	Actuator	Objective	Market / Control
Inertia	0 — 2 s	Synchronous rotors	Limit RoCoF (df/dt)	Physical / Grid-forming
Primary (FCR)	2 — 30 s	Speed governors	Arrest nadir / f_{ss}	Decentralized droop (σ)
Secondary (aFRR)	0.5 — 15 min	Centralized AGC	$\Delta f \to 0$, $\Delta P_{tie} \to 0$	PICASSO / PI loop
Tertiary (mFRR)	15 — 60 min	Peakers, pumps	Restore aFRR	MARI / Manual dispatch
Economic Dispatch	15 min — 1 d	Fleet scheduling	Minimize $\sum C_i$	Merit Order / OPF

Conclusion of Lecture

Core Takeaways:

- Frequency is a global system variable dictated by active power balance.
- Kinetic energy buffers initial shocks; droop controls arrest the fall.
- AGC restores nominal frequency and contracted exchanges without inter-TSO voice calls.
- Economic dispatch and OPF ensure cost efficiency while respecting network transmission bottlenecks.

Next Lecture

Voltage Stability: $P - V$ and $Q - V$ nose curves, reactive power compensation, tap changers, and maximum loadability limits.